

Phase 1: Source-side memory training

(1) Source Engram (trained on corpus C)



Engram Training
next-token prediction
(unsupervised)

Engram Table E^A



Source model and Engram table
are frozen at end of Phase 1.

(2) Input-Space Addressing (tokenizer-agnostic)

Raw tokens from any tokenizer



Canonicalization

$P(\cdot)$: NFKC + lowercase +
accent stripping

Canonical tokens



N-gram Hashing ($\varphi_{n,k}$)

$n = 1 \dots N, k = 1 \dots K$

Memory indices (per head)



(3) Frozen Engram Memory Table Engram Memory Table E shared across models frozen

Head 1 $\square$ $\square$ $\square$... $\square$

Head K $\square$ $\square$ $\square$... $\square$

Slot 0 Slot 1 Slot 2 Slot ($M-1$)

Phase 2: Target-side reader adaption

(4) Target Model + Trainable Reader (Extractor)

Target Model B



Backbone hidden state h_t

Trainable Reader W^B
(learned in Phase 2 only)

Key Projection

W_K^B

Key Vectors

h_t

Value Projection

W_V^B

Value Vectors

v_t

Gate (context-aware)

$$\alpha_t = \sigma\left(\frac{h_t^T \text{RMSNorm}(k_t)}{\sqrt{d_B}}\right)$$

Lookup per head



$\alpha_t \cdot v_t$



Augmented
hidden state